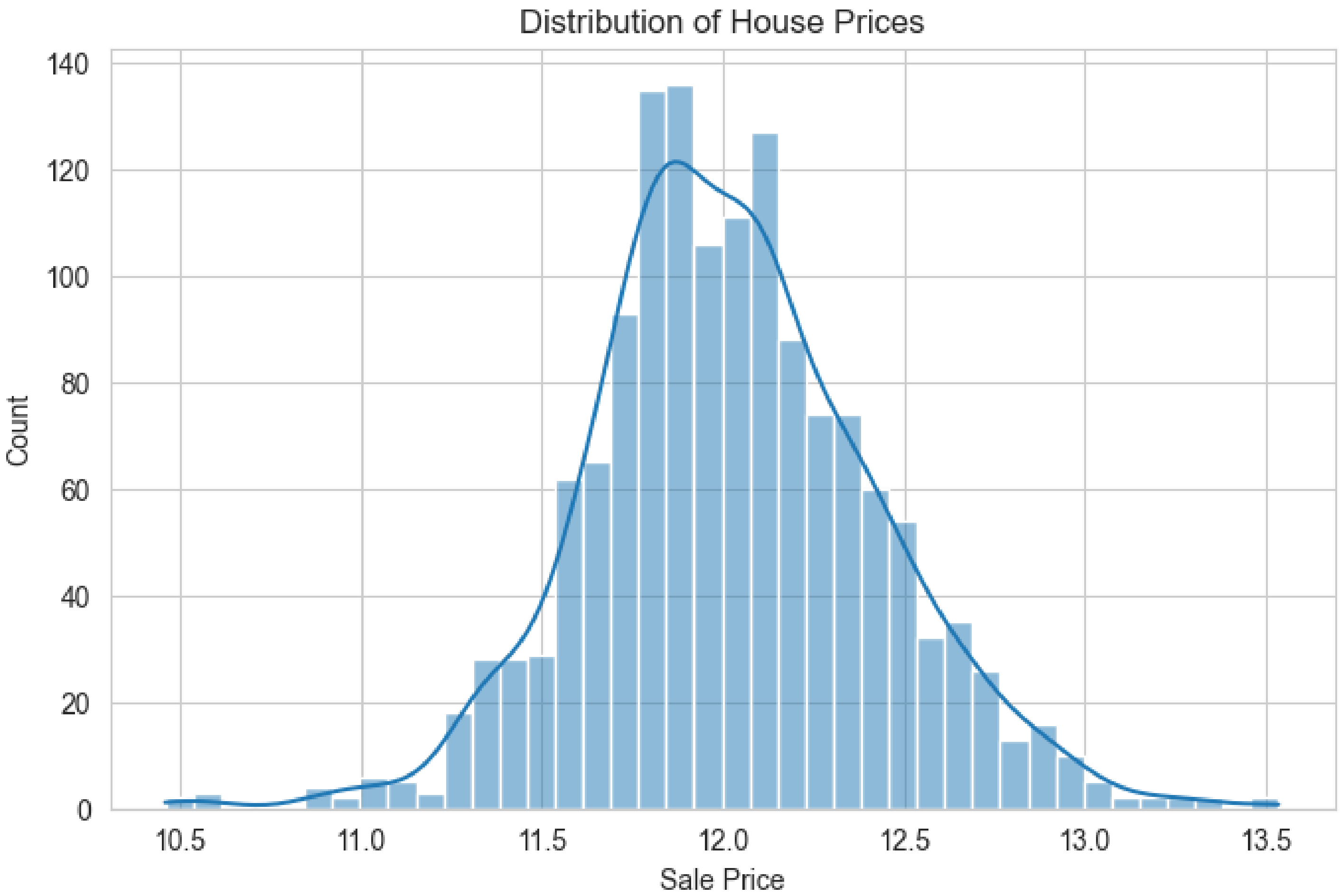
HOUSE PRICE ANALYSIS

Descriptive Statistical Report

Graphs & Plots

Distribution of House Prices



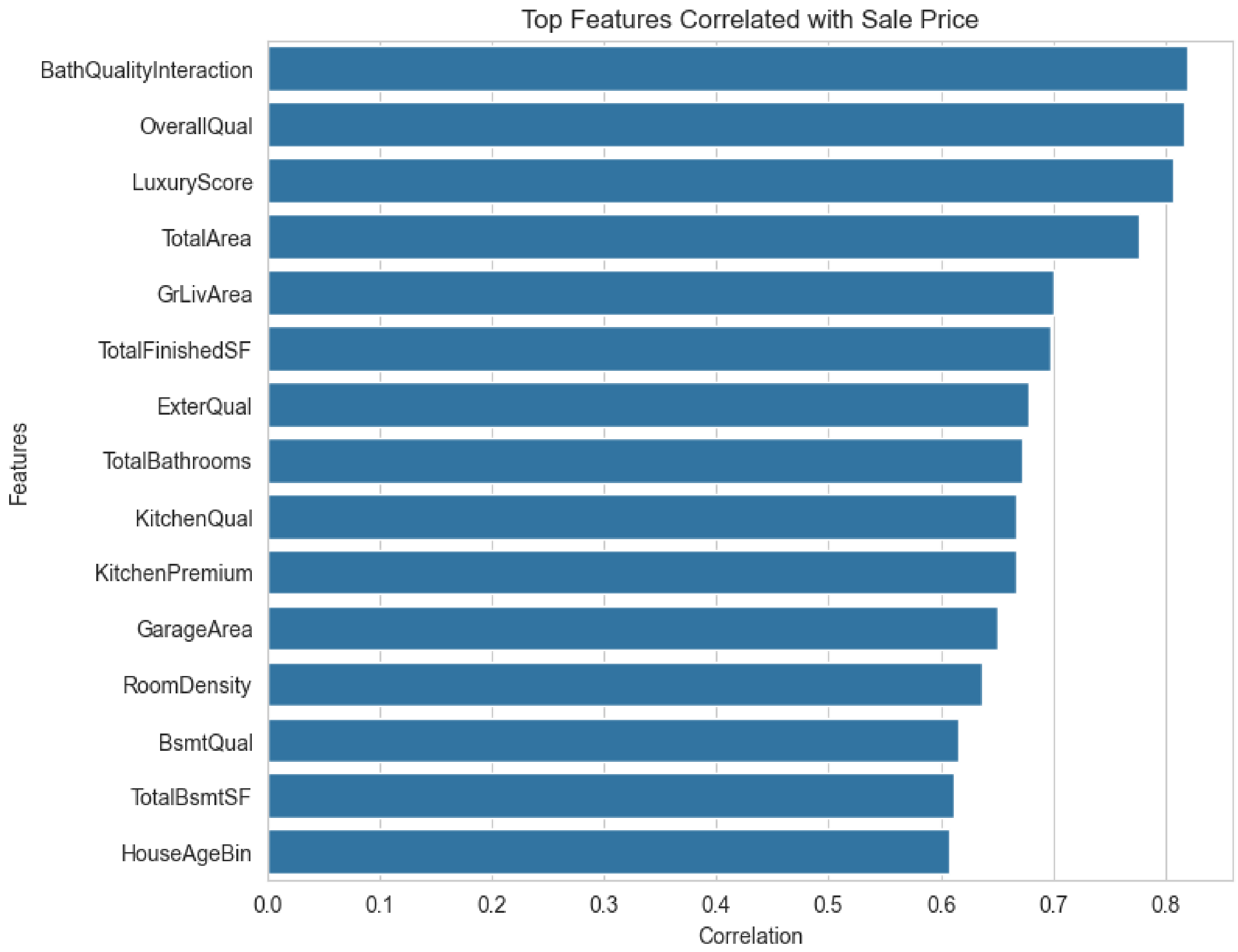
X axis = log(Sale Price)

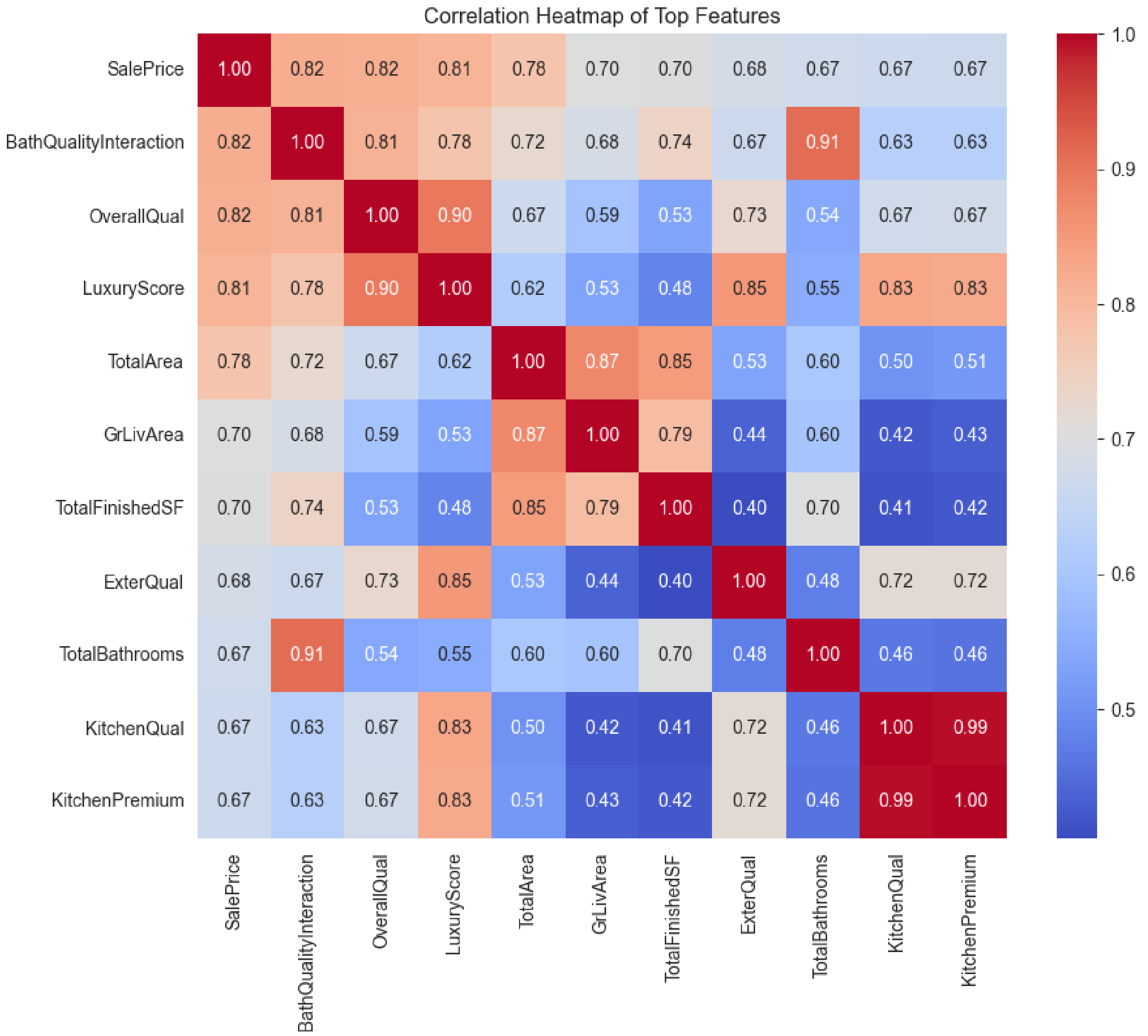
The distribution of house prices is approximately **bell-shaped** and close to a normal distribution, with a slight right skew.
Most houses have **log-transformed sale prices between 11.7 and 12.3**, indicating that the majority of properties fall within this range.
The highest concentration of house prices is around **11.9–12.0**, which represents the most common sale price.
A small number of properties exist at both the lower and higher ends of the distribution, indicating the presence of **potential outliers**.
The slight positive skew suggests there are **more high-priced houses than low-priced houses**, though extreme values are relatively rare.

Correlation with Sale Price

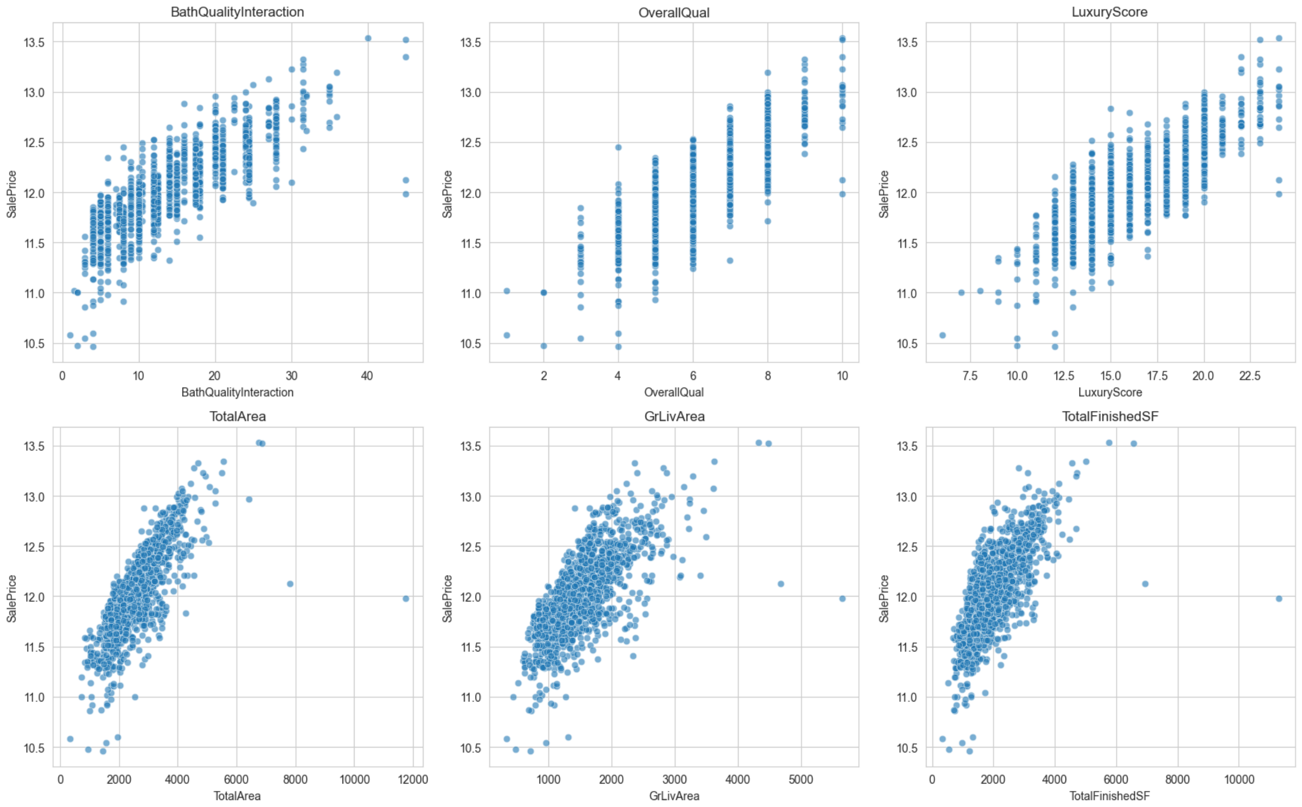
Top 15 features with +ve correlation

BathQualityInteraction	0.819640
OverallQual	0.817185
LuxuryScore	0.807407
TotalArea	0.777297
GrLivArea	0.700927
TotalFinishedSF	0.698492
ExterQual	0.678840
TotalBathrooms	0.673011
KitchenQual	0.667893
KitchenPremium	0.666723
GarageArea	0.650888
RoomDensity	0.636987
BsmtQual	0.615804
TotalBsmtSF	0.612134
HouseAgeBin	0.608107




Correlation Heatmap of Top Features
SalePrice 1.00 0.82 0.82 0.81 0.78 0.70 0.70 0.68 0.67 0.67 0.67
BathQualityInteraction 0.82 1.00 0.81 0.78 0.72 0.68 0.74 0.67 0.91 0.63 0.63
OverallQual 0.82 0.81 1.00 0.90 0.67 0.59 0.53 0.73 0.54 0.67 0.67
LuxuryScore 0.81 0.78 0.90 1.00 0.62 0.53 0.48 0.85 0.55 0.83 0.83
TotalArea 0.78 0.72 0.67 0.62 1.00 0.87 0.85 0.53 0.60 0.50 0.51
GrLivArea 0.70 0.68 0.59 0.53 0.87 1.00 0.79 0.44 0.60 0.42 0.43
TotalFinishedSF 0.70 0.74 0.53 0.48 0.85 0.79 1.00 0.40 0.70 0.41 0.42
ExterQual 0.68 0.67 0.73 0.85 0.53 0.44 0.40 1.00 0.48 0.72 0.72
TotalBathrooms 0.67 0.91 0.54 0.55 0.60 0.60 0.70 0.48 1.00 0.46 0.46
KitchenQual 0.67 0.63 0.67 0.83 0.50 0.42 0.41 0.72 0.46 1.00 0.99
KitchenPremium 0.67 0.63 0.67 0.83 0.51 0.43 0.42 0.72 0.46 0.99 1.00
SalePrice
BathQualityInteraction
OverallQual
LuxuryScore
TotalArea
GrLivArea
TotalFinishedSF
ExterQual
TotalBathrooms
KitchenQual
KitchenPremium
1.0
0.9
0.8
0.7
0.6
0.5

Scatterplots of Most Important Numerical Features



Overall positive relationship: All six features show a positive correlation with SalePrice, indicating that higher values of these features are generally associated with higher house prices.

Strongest predictors: TotalArea, TotalFinishedSF, and GrLivArea exhibit the strongest linear relationships with SalePrice, making them highly influential predictors for house price estimation.

Quality matters: OverallQual, BathQualityInteraction, and LuxuryScore also show strong positive trends, suggesting that better construction quality, bathroom quality, and luxury-related features significantly increase property value.

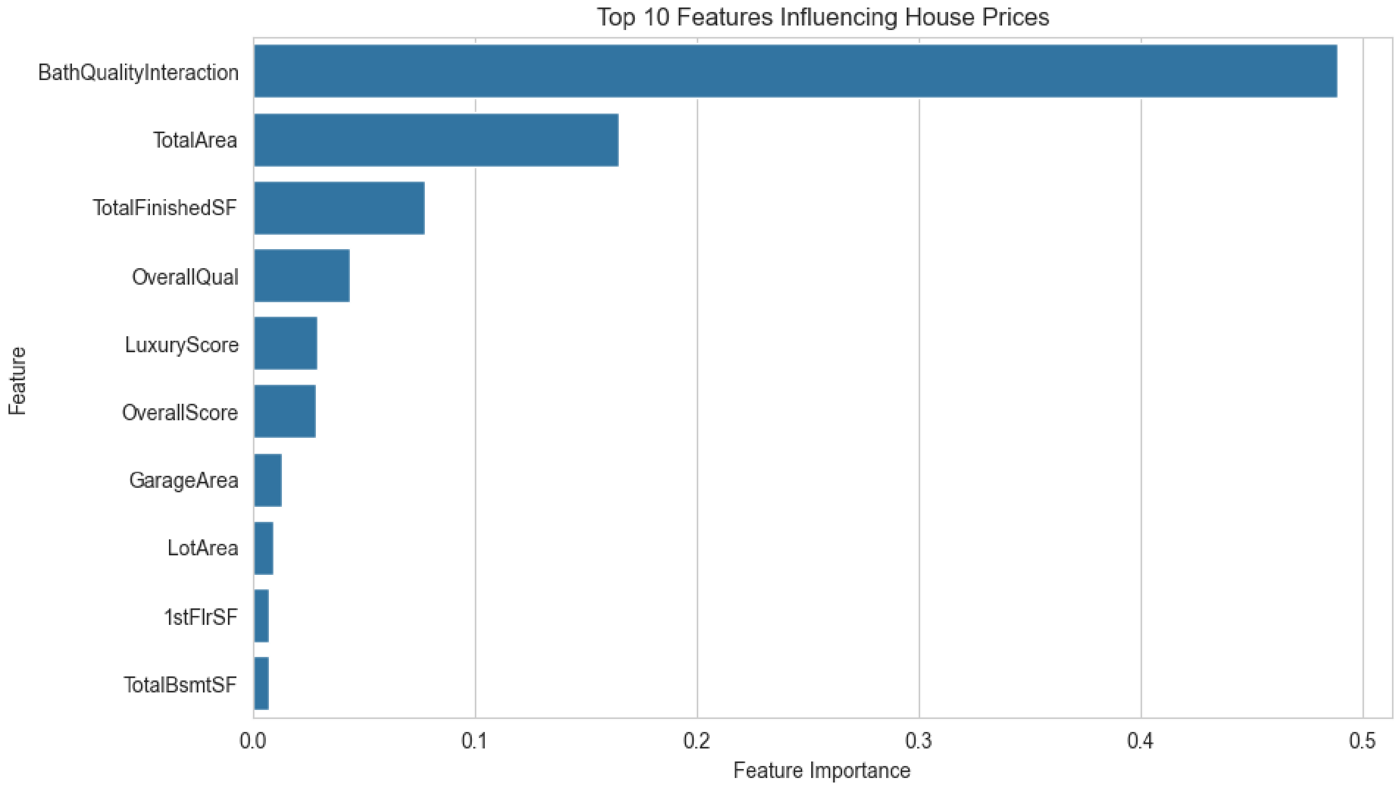
Presence of outliers: A few observations have exceptionally large area values but comparatively lower sale prices, indicating the presence of outliers that may require further investigation or treatment before modelling.

Feature engineering is effective: The engineered features (BathQualityInteraction, LuxuryScore, and TotalArea) demonstrate clear correlations with SalePrice, confirming that feature engineering has improved the representation of factors influencing house prices.

Feature Importance

	Feature	Importance
199	BathQualityInteraction	0.489014
190	TotalArea	0.164675
191	TotalFinishedSF	0.077049
8	OverallQual	0.043814
216	LuxuryScore	0.029052
214	OverallScore	0.028536
40	GarageArea	0.012953
5	LotArea	0.009164
27	1stFlrSF	0.007222
23	TotalBsmtSF	0.006848

Top 10
Most Important Features



BathQualityInteraction is the most influential feature, contributing nearly **50%** of the model's predictive importance, indicating that bathroom quality has a major impact on house prices.

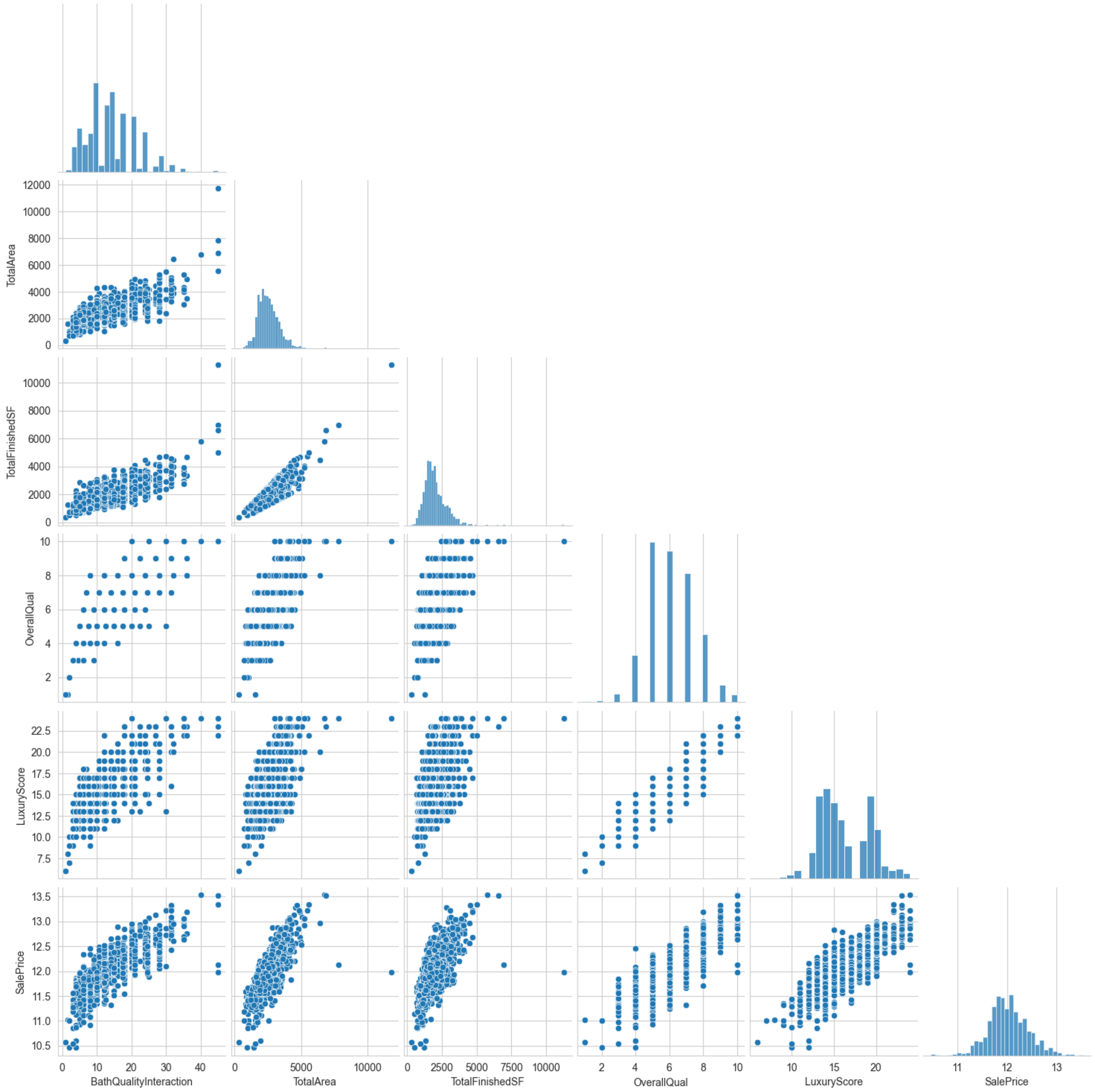
TotalArea is the second most important feature, highlighting that the overall size of the property is a key determinant of its market value.

TotalFinishedSF, **OverallQual**, and **LuxuryScore** also contribute significantly, showing that finished living space, construction quality, and luxury-related features strongly influence sale prices.

raditional features such as **GarageArea**, **LotArea**, **1stFlrSF**, and **TotalBsmtSF** have comparatively lower importance, suggesting they provide additional but limited predictive value.

The results indicate that the **engineered features** (BathQualityInteraction, TotalArea, and LuxuryScore) are among the most influential predictors, demonstrating that feature engineering substantially improved the model's ability to capture the factors affecting house prices.

Pairplot of Top Features



Conclusion

•Performed comprehensive **Exploratory Data Analysis (EDA)** to understand the distribution of house prices, detect outliers, and identify relationships between key variables.

•Applied **feature engineering** to create meaningful variables such as **BathQualityInteraction**, **LuxuryScore**, **TotalArea**, and **TotalFinishedSF**, which significantly improved the representation of property characteristics.

•Correlation analysis revealed that **BathQualityInteraction (0.82)**, **OverallQual (0.82)**, **LuxuryScore (0.81)**, and **TotalArea (0.78)** are the strongest predictors of house prices.

•Feature importance analysis confirmed that the engineered features, particularly **BathQualityInteraction**, contributed the most to the predictive performance of the model.

•Scatter plots and pair plots demonstrated strong positive relationships between house prices and features related to **quality**, **living area**, and **luxury amenities**.

•The target variable (**SalePrice**) exhibited an approximately normal distribution after **log transformation**, making it more suitable for machine learning algorithms.

•Overall, the analysis shows that **property quality, usable living space, and engineered composite features** have a greater influence on house prices than individual structural attributes alone.